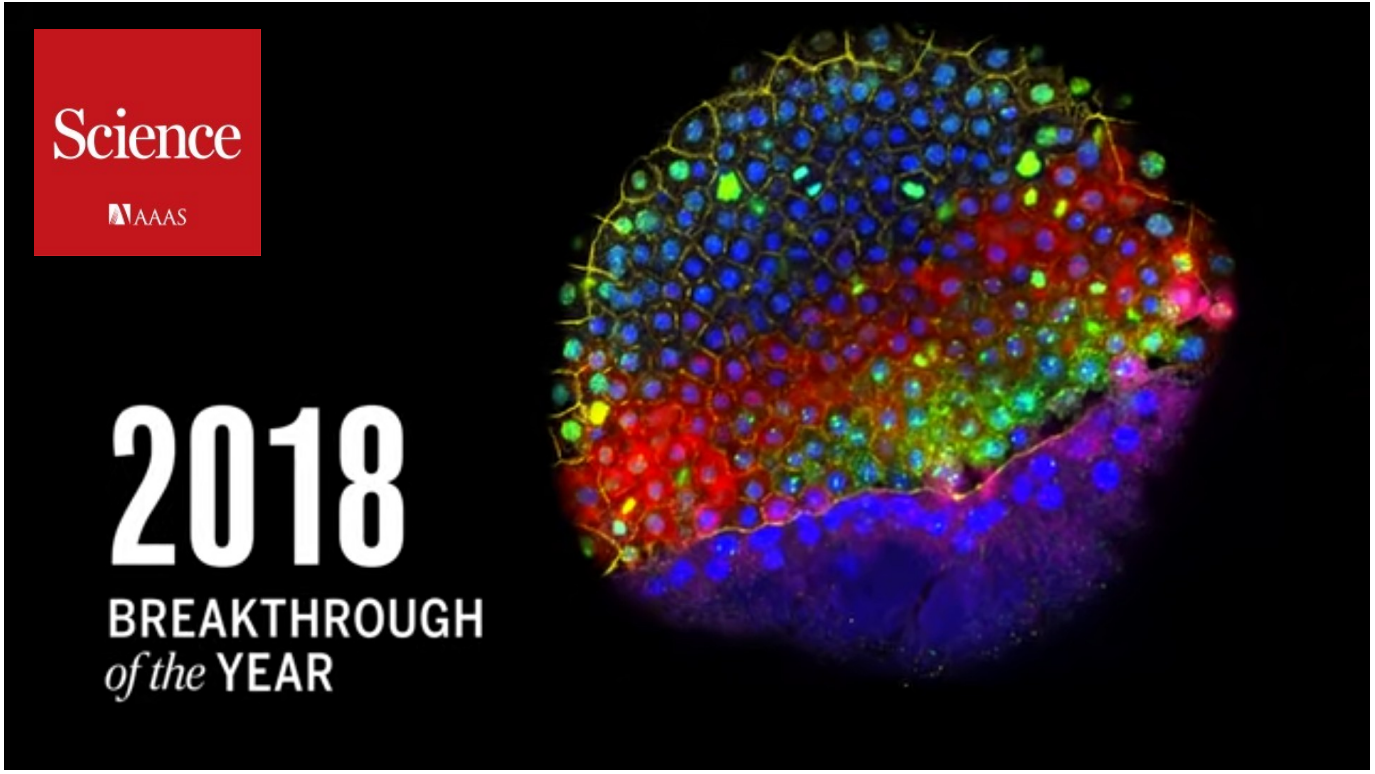
Single Cell RNAseq Analysis

Nikolay Oskolkov, Metabolic Research Group leader at NIRI, Riga, Latvia

PR Stats course, **Session 3: normalization and batch-correction of scRNAseq data**



@NikolayOskolkov

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Personal homepage:
https://nikolay-oskolkov.com

Nature method of the year:
2019 (single cell multi-omics),
2020 (spatial transcriptomics)

Image adapted from Science, 362, 6421, 2018

Session content

Topics we'll cover in this session:

1) Normalization and batch correction

2) Comparison of different normalization methods

3) Comparison of different batch correction methods

Removing technical variation:

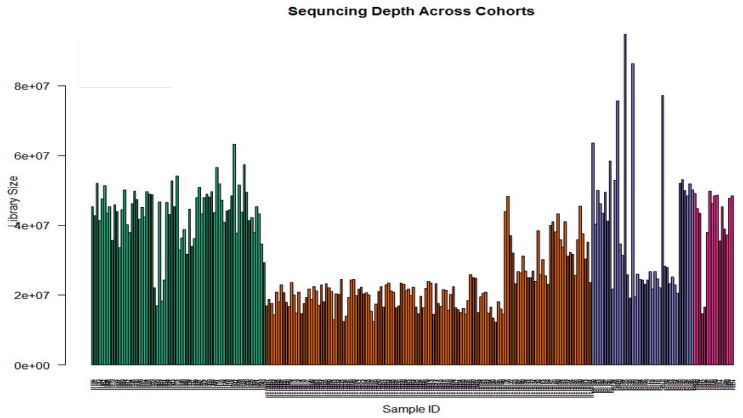
1) Normalization

2) Batch correction

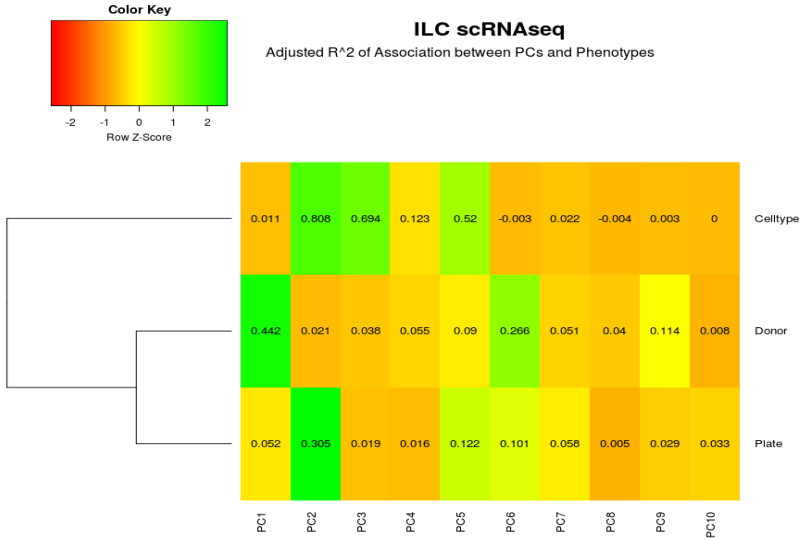
Batch correction:
genome-wide and gene-specific

How to Detect Technical Variation

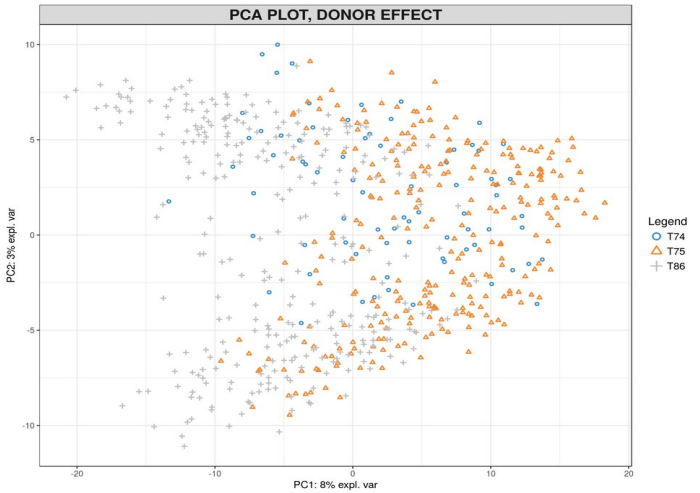
Difference in sequencing depth:



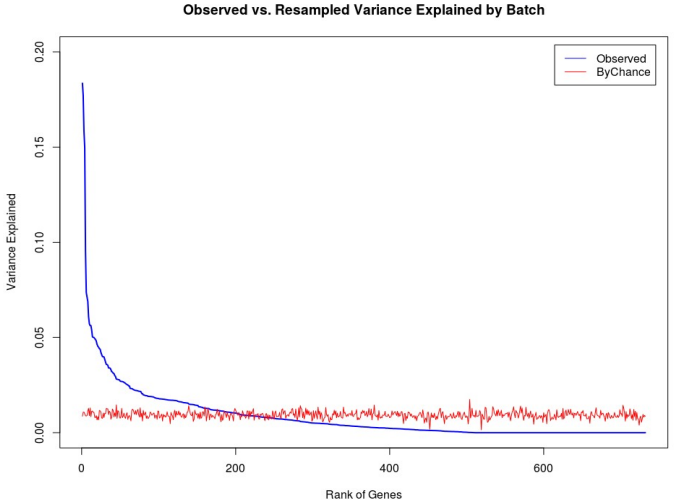
Genome-Wide Batch-effects:



Genome-Wide Batch-effects:



Per-Gene Batch-effects:



Batch Correction: Linear vs. Non-Linear

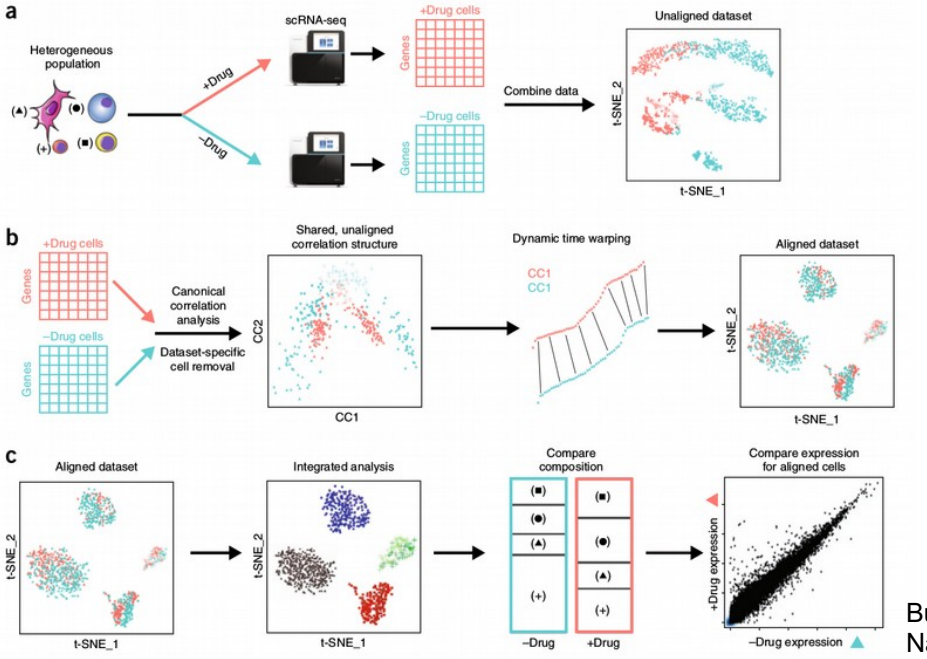
Biostatistics (2007), **8**, 1, pp. 118–127
doi:10.1093/biostatistics/kxj037
Advance Access publication on April 21, 2006

Limma and Combat:
gene_expr ~ cell_type + batch

Adjusting batch effects in microarray expression data using empirical Bayes methods

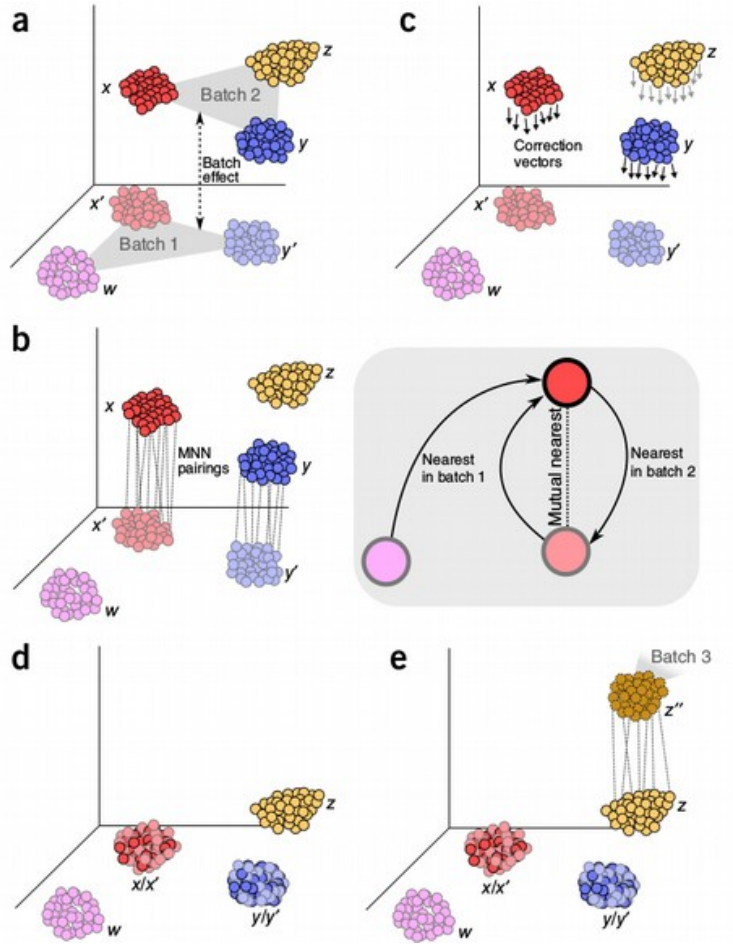
W. EVAN JOHNSON, CHENG LI*
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Department of Genetics and Complex Diseases, Harvard School of Public Health, Boston, MA, USA

Seurat: CCA + DTW



Butler et al.,
Nature Biotechnology 2018

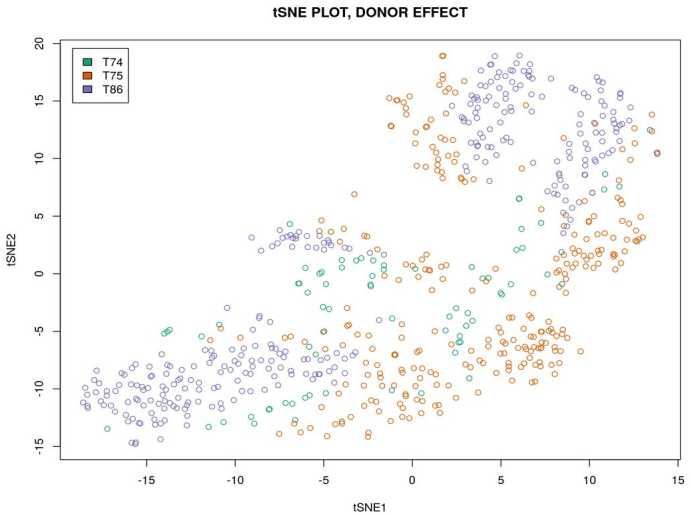
Mutual Nearest Neighbor (MNN)



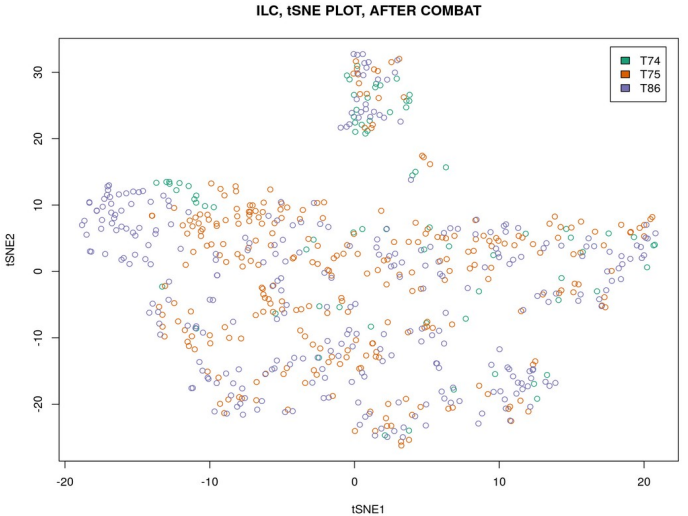
Haghverdi et al.,
Nature Biotechnology 2018

Batch Correction: Algorithm Comparison

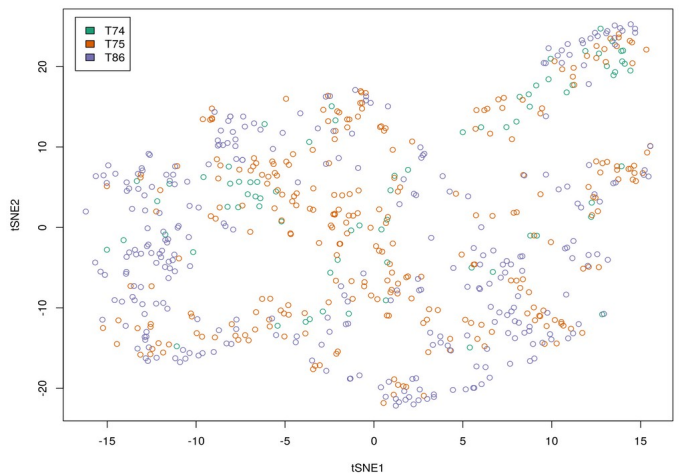
UNCORRECTED



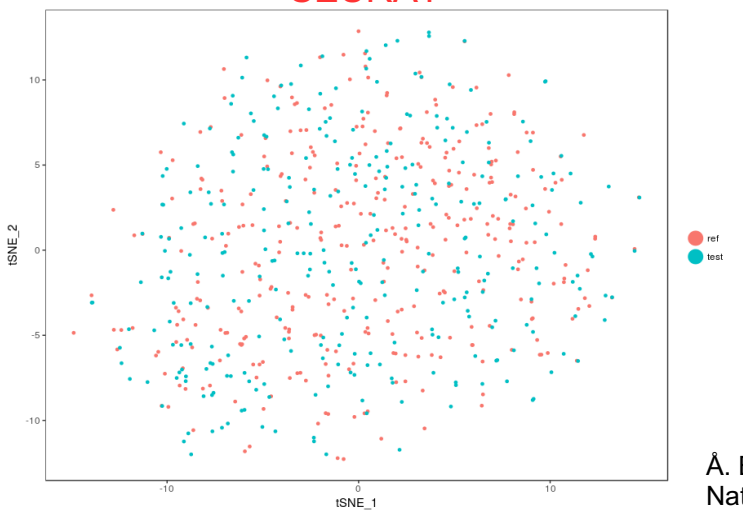
COMBAT



MNN



SEURAT



Å. Björklund et al.,
Nature Immunology 2016

Normalization:
bulk RNAseq and scRNAseq-specific

RPKMs (FPKMs)

RPKM normalization is an extension of so-called library size normalization

Library size normalization: scaling such that library size is equal between all libraries

$$RPKM = \frac{10^9 C}{NL}$$

where:

C = number of reads that overlap a given gene

N = library size

L = gene length

Disadvantage: forced equalizing library sizes might eliminate true biological variation

Traditional Normalization: DESeq, TMM

Build reference library based on geometric mean of all libraries, calculate size factors as ratios against the reference library

$$\begin{array}{c} \\ g_1 \\ g_2 \\ g_n \end{array}\begin{array}{c} \begin{array}{ccc} s_1 & s_2 & s_k \end{array} \\ \begin{bmatrix} c_{11} & c_{12} & c_{1k} \\ c_{21} & c_{22} & c_{2k} \\ c_{n1} & c_{n2} & c_{nk} \end{bmatrix} \end{array} \begin{bmatrix} 1/g_{1r} & 1/g_{2r} & 1/g_{nr} \end{bmatrix} \longrightarrow \begin{array}{c} \\ g_1 \\ g_2 \\ g_n \end{array}\begin{array}{c} \begin{array}{ccc} s_1 & s_2 & s_k \end{array} \\ \begin{bmatrix} r_{11} & r_{12} & r_{1k} \\ r_{21} & r_{22} & r_{2k} \\ r_{n1} & r_{n2} & r_{nk} \end{bmatrix} \end{array}$$

estimate the relative depth of the library

$S_1 = \text{median}(r_{x1})$

take the mean for each row to obtain a reference sample

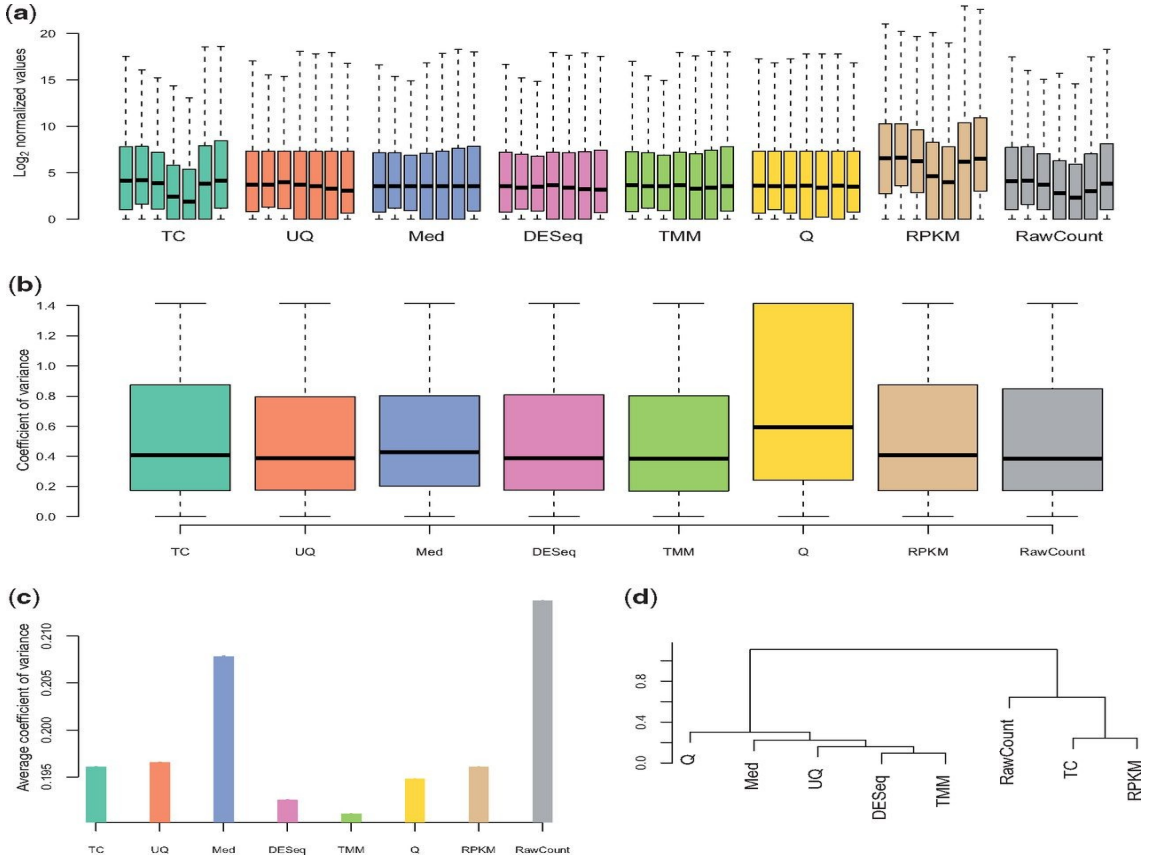
$$\begin{bmatrix} g_{1r} \\ g_{2r} \\ g_{nr} \end{bmatrix}$$

estimate the depth ratio for each gene

$$\begin{array}{c} \\ g_1 \\ g_2 \\ g_n \end{array}\begin{array}{c} \begin{array}{ccc} s_1 & s_2 & s_k \end{array} \\ \begin{bmatrix} c_{11} & c_{12} & c_{1k} \\ c_{21} & c_{22} & c_{2k} \\ c_{n1} & c_{n2} & c_{nk} \end{bmatrix} \end{array} \begin{bmatrix} S_1 \\ S_2 \\ S_k \end{bmatrix}$$

PROBLEM: scRNAseq count matrix has a lot of zeroes, hence size factors might be diverging!

TMM and DESeq Minimize Technical Variation



Marie-Agnès Dillies et al. Brief Bioinform 2012;bib.bbs046

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Briefings in Bioinformatics

Deconvolution Normalization

Lun *et al. Genome Biology* (2016) 17:75
DOI 10.1186/s13059-016-0947-7

Genome Biology

METHOD Open Access

Pooling across cells to normalize single-cell RNA sequencing data with many zero counts

Aaron T. L. Lun[1*], Karsten Bach[2] and John C. Marioni[1,2,3*]

Abstract

Normalization of single-cell RNA sequencing data is necessary to eliminate cell-specific biases prior to downstream analyses. However, this is not straightforward for noisy single-cell data where many counts are zero. We present a novel approach where expression values are summed across pools of cells, and the summed values are used for normalization. Pool-based size factors are then deconvolved to yield cell-based factors. Our deconvolution approach outperforms existing methods for accurate normalization of cell-specific biases in simulated data. Similar behavior is observed in real data, where deconvolution improves the relevance of results of downstream analyses.

Keywords: Single-cell RNA-seq, Normalization, Differential expression

Background

Single-cell RNA sequencing (scRNA-seq) is a powerful technique that allows researchers to characterize the gene expression profile of single cells. From each cell, mRNA is isolated and reverse-transcribed into cDNA, which is amplified and subjected to massively parallel sequencing [1]. The sequencing reads are mapped to a reference genome, such that the number of reads mapped to each gene can be used to quantify its expression. Alternatively, transcript molecules can be counted directly using unique molecular identifiers (UMIs) [2]. Count data can be analyzed to identify new cell subtypes and to detect highly variable or differentially expressed (DE) genes between cell subpopulations. This type of single-cell resolution is not possible with bulk RNA sequencing of cellular populations. However, the downside is that the counts often contain high levels of technical noise with many dropouts, i.e., zero or near-zero values. This is due to the presence of low amounts of RNA per cell, which decreases the efficiency with which transcripts can be captured and processed prior to sequencing. Moreover, the capture efficiency often varies from cell to cell, such that counts cannot be directly compared between cells.

Normalization of the scRNA-seq counts is a critical step that corrects for cell-to-cell differences in capture efficiency, sequencing depth, and other technical confounders. This ensures that downstream comparisons of relative expression between cells are valid. Two broad classes of methods for scaling normalization are available: those using spike-in RNA sets and those using the counts from the profiled cellular RNA. In the former, the same quantity of spike-in RNA is added to each cell prior to library preparation [1]. Any difference in the coverage of the spike-in transcripts must be caused by differences in capture efficiency, amplification bias, or sequencing depth between cells. Normalization is then performed by scaling the counts to equalize spike-in coverage between cells. For the methods using cellular counts, the assumption is that most genes are not DE across the sampled cells. Counts are scaled so that there is, on average, no fold-difference in expression between cells for the majority of genes. This is the underlying concept of commonly used methods such as DESeq [3] and trimmed mean of M values (TMM) normalization [4]. An even simpler approach involves scaling the counts to remove differences in library sizes between cells, i.e., library size normalization.

The type of normalization that can be used depends on the characteristics of the data set. In some cases, spike-in

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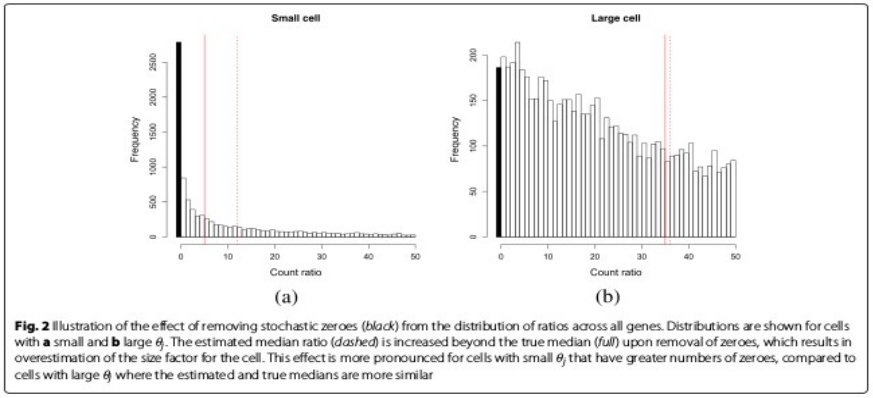


Fig. 2 Illustration of the effect of removing stochastic zeroes (*black*) from the distribution of ratios across all genes. Distributions are shown for cells with **a** small and **b** large θ_j. The estimated median ratio (*dashed*) is increased beyond the true median (*full*) upon removal of zeroes, which results in overestimation of the size factor for the cell. This effect is more pronounced for cells with small θ_j that have greater numbers of zeroes, compared to cells with large θ_j where the estimated and true medians are more similar

of an arbitrary set of cells $\mathcal{S}_k$. Define V_{ik} as the sum of Z_{ij} across all cells in $\mathcal{S}_k$, which has an expectation of

$$E(V_{ik}) = \lambda_{i0} \sum_{j \in \mathcal{S}_k} \theta_j t_j^{-1}.$$

The observed values of V_{ik} across all genes constitute an overall expression profile for the pool of cells corresponding to $\mathcal{S}_k$. Also define U_i as the mean of Z_{ij} across all N cells in the entire data set, which has an expectation of

$$E(U_i) = \lambda_{i0} N^{-1} \sum_{j \in \mathcal{S}_0} \theta_j t_j^{-1}$$

where $\mathcal{S}_0$ refers to the set of all cells in the data set. The observed values of U_i across all genes represent the expression profile for an averaged reference pseudo-cell.

The cell pool k is then normalized against this reference pseudo-cell. Define R_{ik} as the ratio of V_{ik} to U_i for the non-DE gene i. The expectation of R_{ik} represents the true size factor for the pooled cells in $\mathcal{S}_k$, and is written as

$$E(R_{ik}) \approx \frac{E(V_{ik})}{E(U_i)} = \frac{\sum_{\mathcal{S}_k} \theta_j t_j^{-1}}{N^{-1} \sum_{\mathcal{S}_0} \theta_j t_j^{-1}} = \frac{\sum_{\mathcal{S}_k} \theta_j t_j^{-1}}{C} \quad (1)$$

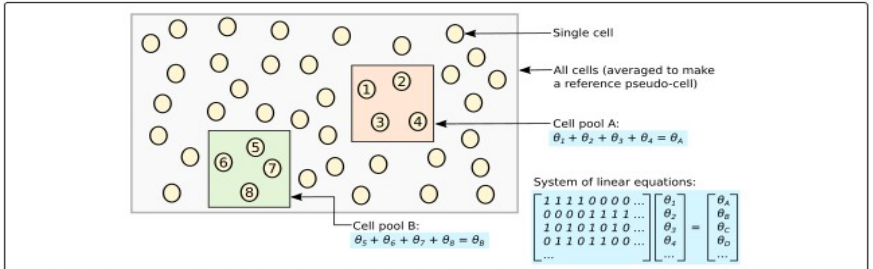


Fig. 3 Schematic of the deconvolution method. All cells in the data set are averaged to make a reference pseudo-cell. Expression values for cells in pool A are summed together and normalized against the reference to yield a pool-based size factor θ_A. This is equal to the sum of the cell-based factors θ_j for cells $j = 1$–4 and can be used to formulate a linear equation. (For simplicity, the t_j term is assumed to be unity here.) Repeating this for multiple pools (e.g., pool B) leads to the construction of a linear system that can be solved to estimate θ_j for each cell j

Benchmarking: Deconvolution Method Performs Best

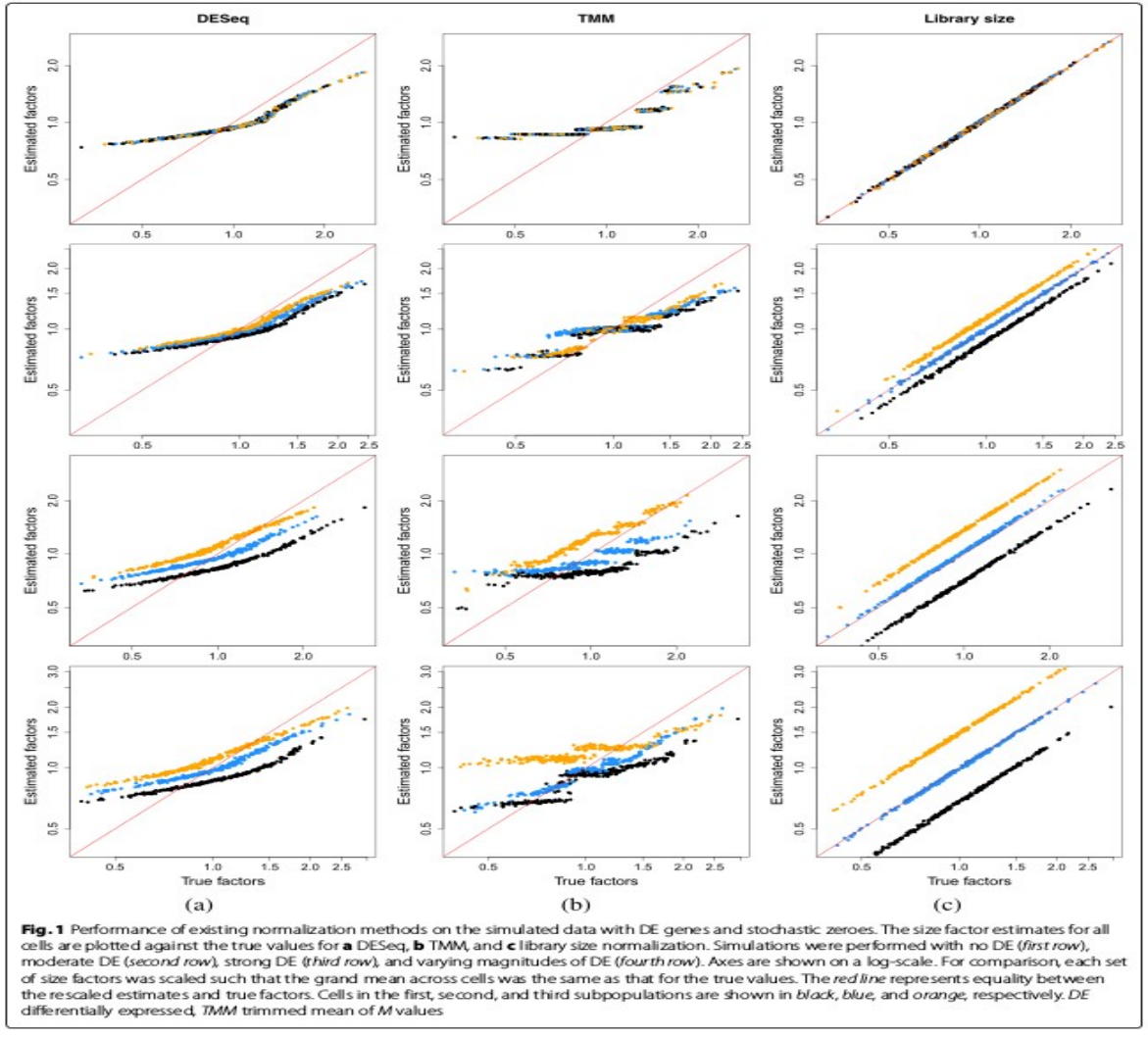
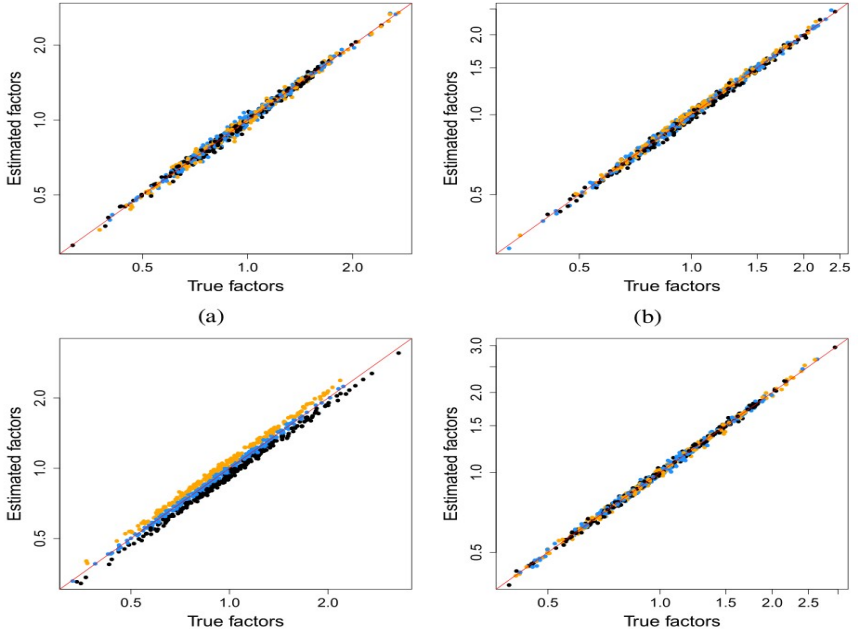
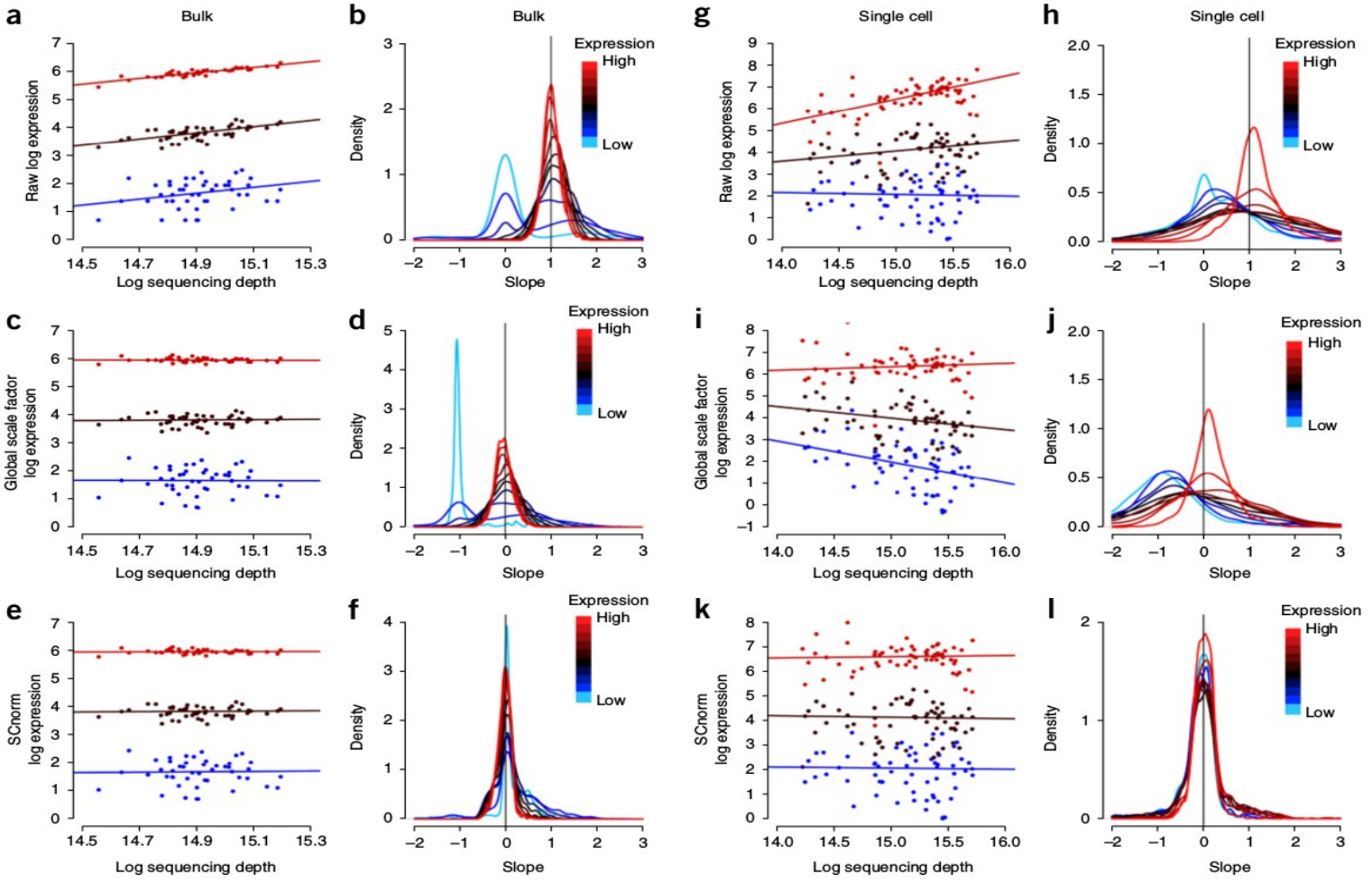


Fig. 1 Performance of existing normalization methods on the simulated data with DE genes and stochastic zeroes. The size factor estimates for all cells are plotted against the true values for **a** DESeq, **b** TMM, and **c** library size normalization. Simulations were performed with no DE (*first row*), moderate DE (*second row*), strong DE (*third row*), and varying magnitudes of DE (*fourth row*). Axes are shown on a log-scale. For comparison, each set of size factors was scaled such that the grand mean across cells was the same as that for the true values. The *red line* represents equality between the rescaled estimates and true factors. Cells in the first, second, and third subpopulations are shown in *black*, *blue*, and *orange*, respectively. *DE* differentially expressed, *TMM* trimmed mean of *M* values

Deconvolution



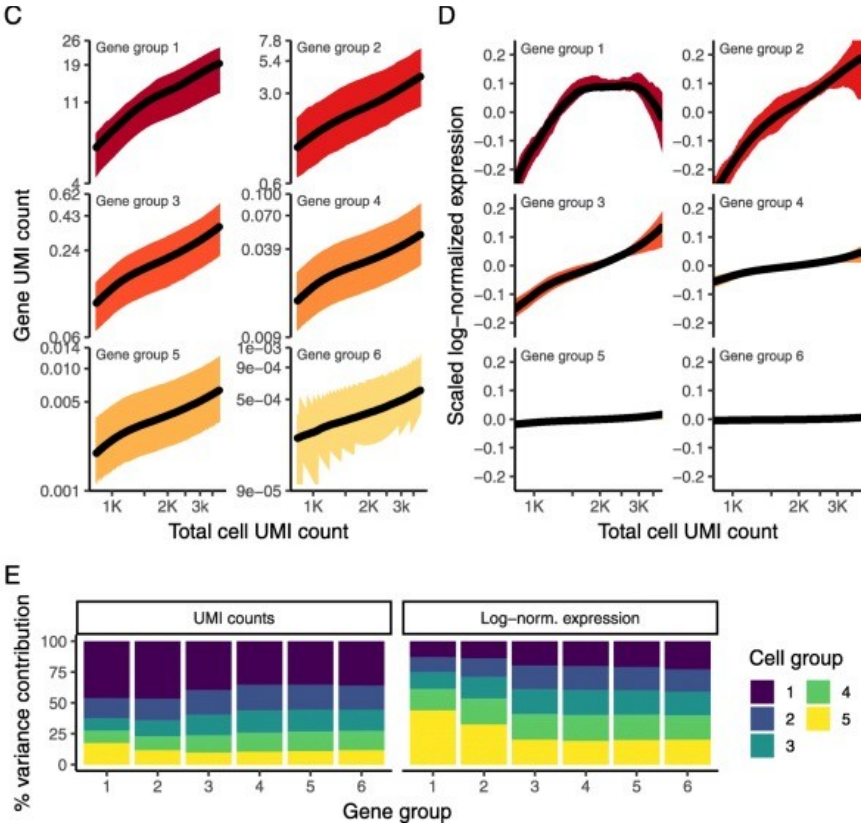
SCNORM Normalization

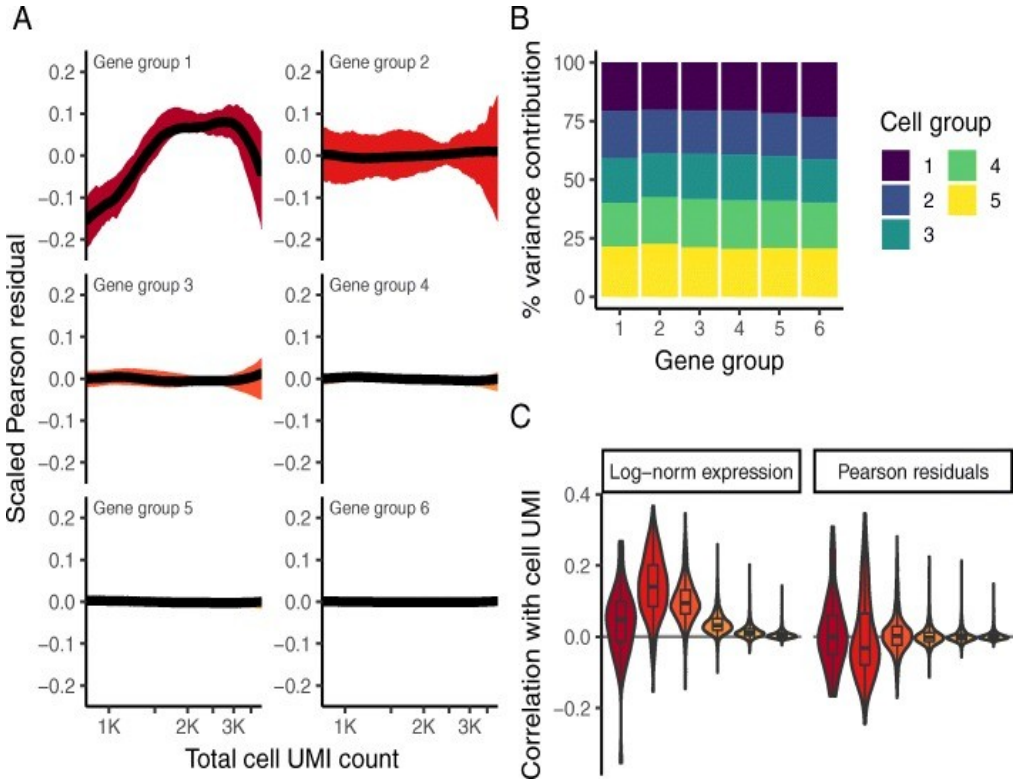


Individual size factor per cell per group of genes

R. Bacher et al.,
Nature Methods 2017

SCTransform Normalization

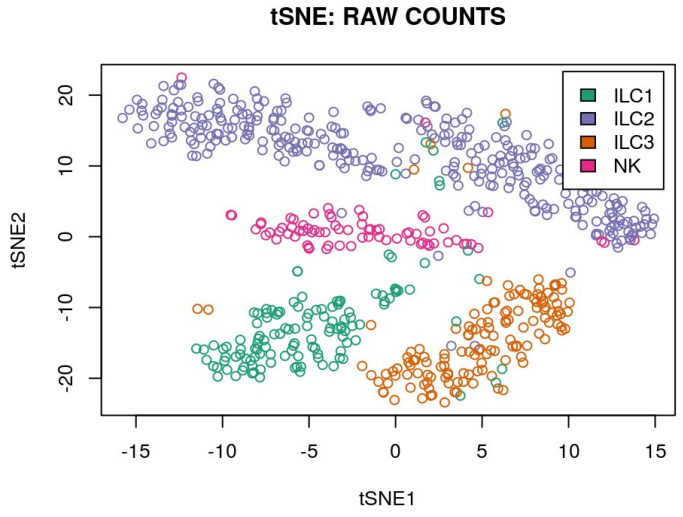


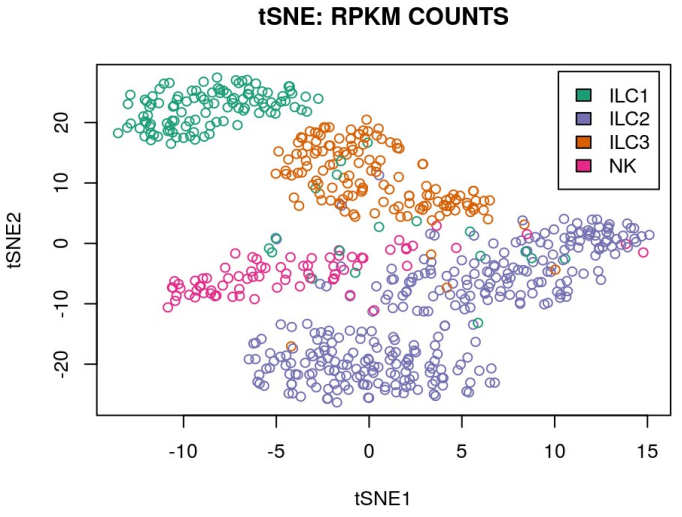


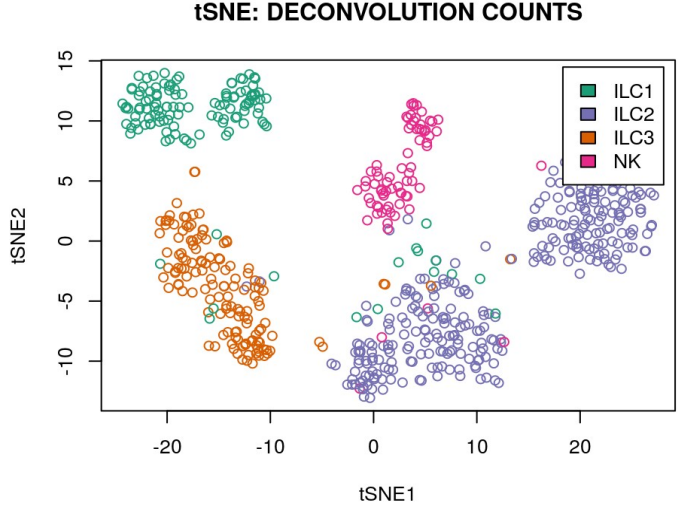
Individual size factor per cell per group of genes

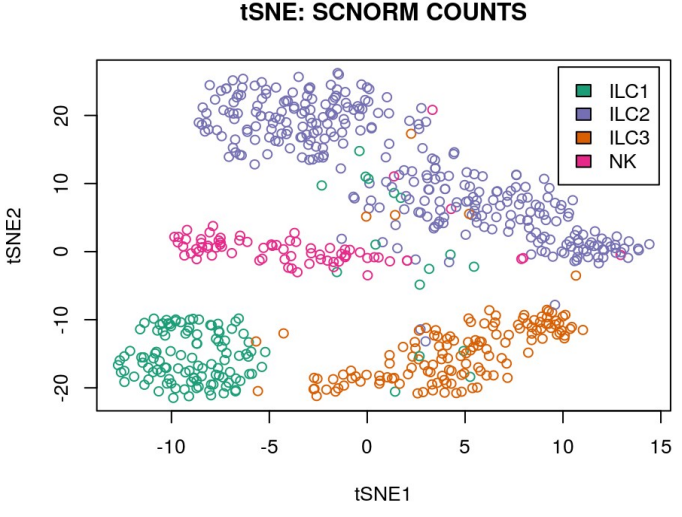
C. Hafemeister et al.,
Genome Biology 2019

Normalization: Algorithm Comparison

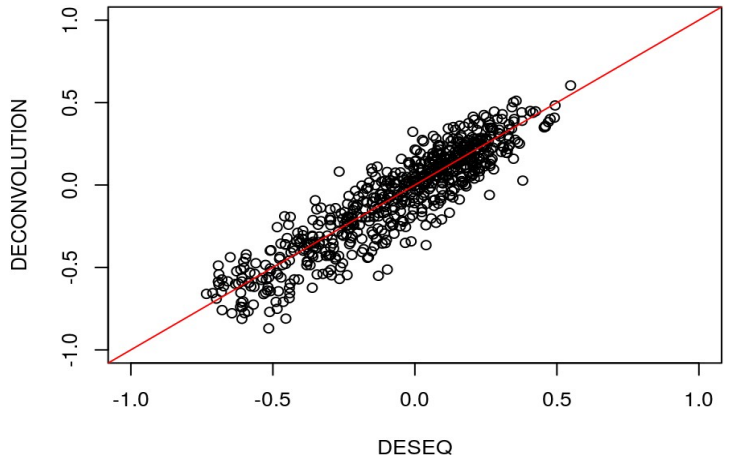


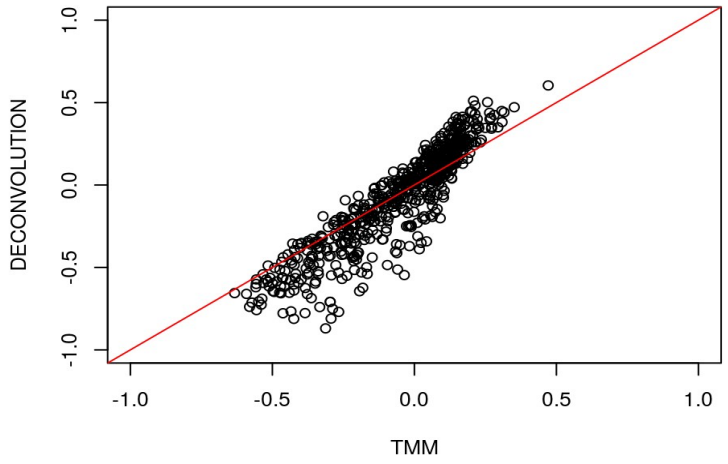


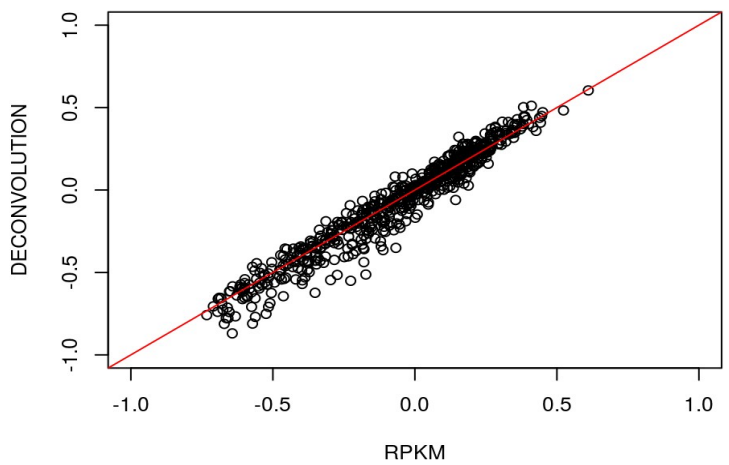


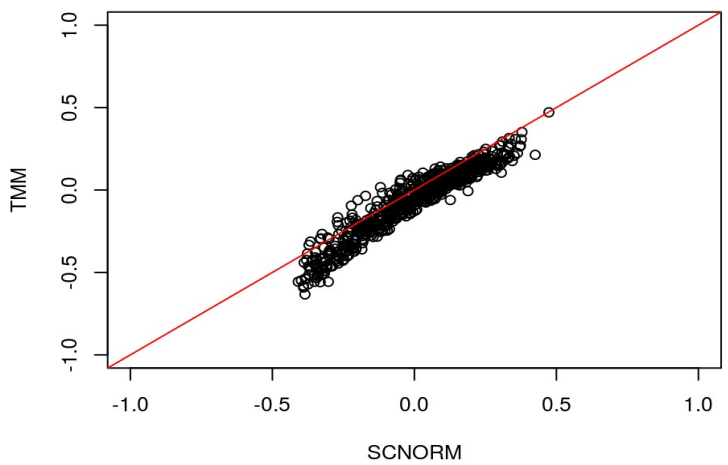


Normalization: Size Factors Comparison









Session summary

Take home messages of the session:

1) Normalization methods have largely comparable effects in practice

2) scNorm and scTransform are recommended for large scRNAseq data

3) Combat corrects well for batch-effects on small scRNAseq data

4) MNN and Seurat CCA are recommended for large nonlinear scRNAseq data

National Bioinformatics Infrastructure Sweden (NBIS)



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Vetenskapsrådet

Lunds Universitet

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